LAPORAN PRAKTIKUM

SISTEM OPERASI

PERTEMUAN 5

(Eksplorasi Proses Xinu)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

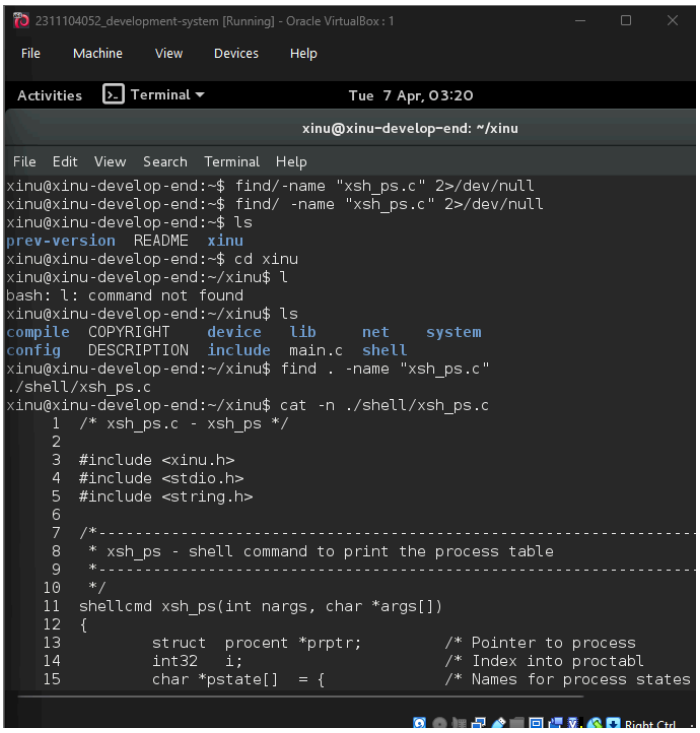
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

Jurnal

1. Jawablah pertanyaan berikut ini:
   a. Berapa banyaknya maksimum proses yang ada pada Xinu?
   = Maksimum proses pada Xinu adalah 8 proses (NPROC = 8). Ini didefinisikan sebagai ukuran array proctab[] pada process.h. Nilai ini bisa berbeda tergantung versi Xinu, namun standarnya adalah 8.
   b. Berapa maksimal panjang nama suatu proses pada Xinu?
   =Maksimal panjang nama proses adalah 16 karakter (PNMLEN = 16). Ini terlihat dari deklarasi char prname[PNMLEN] pada struct procent.
   c. Berapa nilai prioritas awal pada saat proses dibuat?
   =Nilai prioritas awal (default) saat proses dibuat adalah 20 (INITPRIO = 20). Ini adalah nilai default yang digunakan fungsi create() saat membuat proses baru.
   d. Ada berapa total state pada Xinu? Sebutkan!
   =PR_FREE (slot kosong/tidak digunakan), PR_CURR (proses sedang berjalan), PR_READY (proses siap dijalankan), PR_RECV (proses menunggu pesan), dan PR_SUSP (proses ditangguhkan/suspended).
2. Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!



2311104052_development-system [Running] - Oracle VirtualBox : 1

File Machine View Devices Help

Activities Terminal Tue 7 Apr, 03:24

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

GNU nano 2.2.6 File: ./shell/xsh_ps.c

```
        /* Output information for each process */

        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen);
        }
        return 0;
}
```

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2311104052_development-system [Running] - Oracle VirtualBox : 1

File Machine View Devices Help

Activities Terminal Tue 7 Apr, 03:26

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

GNU nano 2.2.6 File: ./shell/xsh_ps.c

```
        /* Output information for each process */
        /*Izzaty zahara Br Barus 2311104052*/
        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen);
        }
        return 0;
}
```

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^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To

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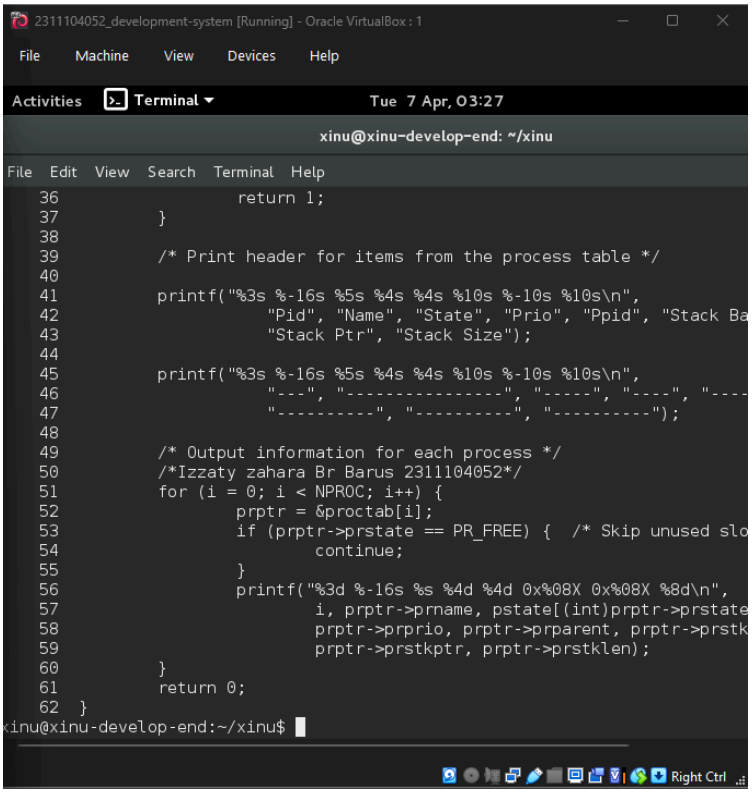

3. Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

```
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
--- ---------------- ----- ---- ---- ---------- ---------- ---------- --- -------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFF04     8192   1       4
  1 rdsproc          susp   200    0 0x005FBFFC 0x005FBFC8    16384   0       0
  2 ipout            wait   500    0 0x005F7FFC 0x005F7F40     8192   0       0
  3 netin            wait   500    0 0x005F5FFC 0x005F5EE0     8192   0       0
  5 Main process     recv    20    4 0x005E3FFC 0x005E3F64    65536   0       0
  6 shell            recv    50    5 0x005D3FFC 0x005D3C7C     8192   0       0
  7 ps               curr    20    6 0x005F3FFC 0x005F3DC8     8192   0       0
```

Kolom Msg adalah banyaknya pesan yang ada dalam proses.

Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah `$ps`. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.
- Screenshot source kode dan output akhir hasil modifikasi

2311104052_development-system [Running] - Oracle VirtualBox : 1

File Machine View Devices Help

Activities Terminal Tue 7 Apr, 03:33

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

GNU nano 2.2.6 File: ./shell/xsh_ps.c

```
        }

        /* Print header for items from the process table */

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------");

        /* Output information for each process */
        /*Izzaty zahara Br Barus 2311104052*/
        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen);
```

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^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To

Right Ctrl

Activities Terminal Tue 7 Apr, 03:42

xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help

GNU nano 2.2.6 File: ./shell/xsh_ps.c Modified

```
        }

        /* Print header for items from the process table */

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %-3s %-7s\n",
                   "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
                   "Stack Ptr", "Stack Size", "Msg", "Content");

        printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %-3s %-7s\n",
                   "---", "----------------", "-----", "----", "----",
                   "----------", "----------", "----------", "---", "-------");

        /* Output information for each process */
        /*Izzaty zahara Br Barus 2311104052*/
        for (i = 0; i < NPROC; i++) {
                prptr = &proctab[i];
                if (prptr->prstate == PR_FREE) {  /* Skip unused slots  */
                        continue;
                }
                printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %7d\n ",
                        i, prptr->prname, pstate[(int)prptr->prstate],
                        prptr->prprio, prptr->prparent, prptr->prstkbase,
                        prptr->prstkptr, prptr->prstklen, prptr->prhasmsg, prptr->prms$
```

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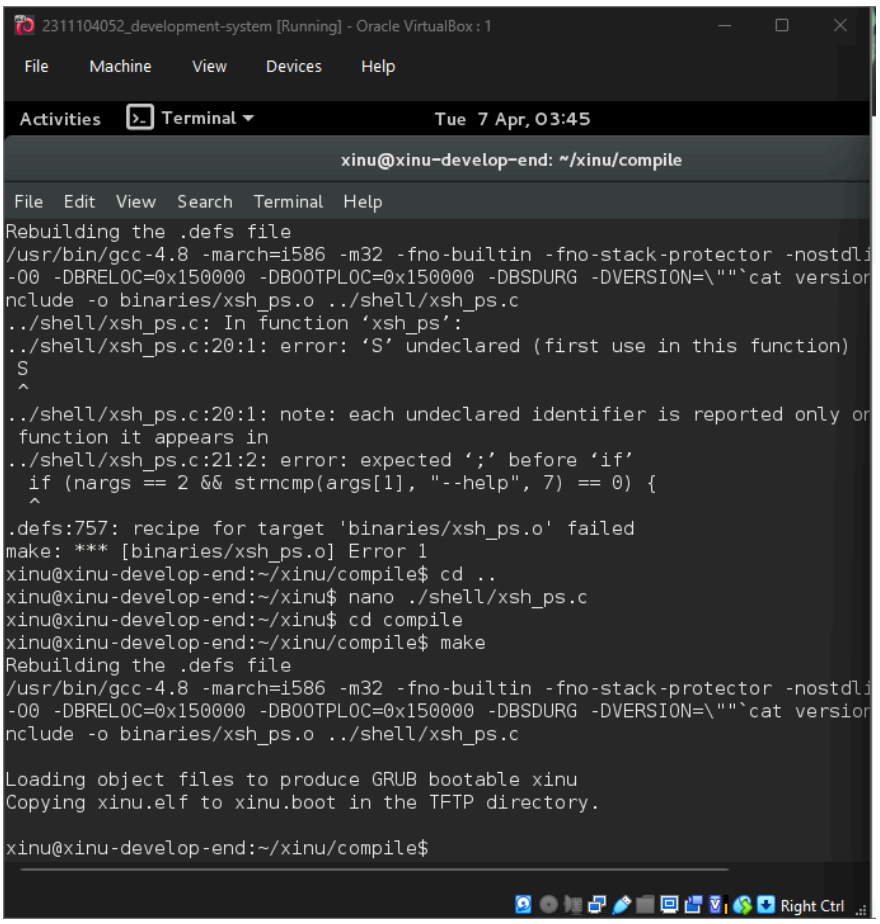
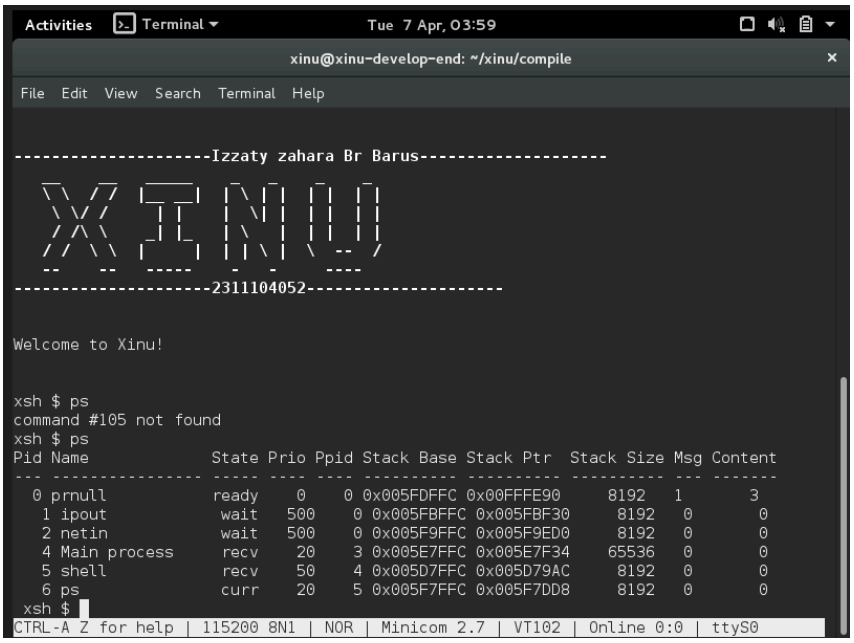
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell





4. Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit. Langkah pengerjaan:
   • Modifikasi source code pada file xsh_uptime.c
   • Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
   • Jalankan Backend VM
   • Setelah sistem berjalan, jalankan perintah $uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan

• Screenshot source kode dan output akhir hasil modifikasi

```
Activities  Terminal ▾        Tue 7 Apr, 04:02
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Tabs  Help
xinu@xinu-develop-end: ~/xinu/compile        xinu@xinu-develop-end: ~/xinu
xinu@xinu-develop-end:~/xinu$ find . -name "xsh_uptime.c"
./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ cat -n ./shell/xsh_uptime.c
     1  /* xsh_uptime.c - xsh_uptime */
     2
     3  #include <xinu.h>
     4  #include <stdio.h>
     5  #include <string.h>
     6
     7  /*------------------------------------------------------------------------
     8   * xsh_uptime - shell to print the time the system has been up
     9   *------------------------------------------------------------------------
    10   */
    11  shellcmd xsh_uptime(int nargs, char *args[])
    12  {
    13          uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
    14                                          /*  seconds since system boot   */
    15          uint32  secperday = 86400;      /* Seconds in a day             */
    16          uint32  secperhr  =  3600;      /* Seconds in an hour           */
    17          uint32  secpermin =    60;      /* Seconds in a minute          */
    18
    19          /* For argument '--help', emit help about the 'uptime' command  */
    20
    21          if (nargs == 2 && strncmp(args[1], "--help", 7) == 0) {
    22                  printf("Use: %s\n\n", args[0]);
```

```
2311104052_development-system [Running] - Oracle VirtualBox : 1
File  Machine  View  Devices  Help
Activities  Terminal ▾        Tue 7 Apr, 04:04
xinu@xinu-develop-end: ~/xinu
File  Edit  View  Search  Terminal  Tabs  Help
xinu@xinu-develop-end: ~/xinu/compile        xinu@xinu-develop-end: ~/xir
  GNU nano 2.2.6              File: ./shell/xsh_uptime.c

/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*------------------------------------------------------------------------
 * xsh_uptime - shell to print the time the system has been up
 *------------------------------------------------------------------------
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
        /*Izzaty zahara BR Barus - 2311104052 */
        uint32  days, hrs, mins, secs;  /* Days, hours, minutes,  and   */
                                        /*  seconds since system boot   */
        uint32  secperday = 86400;      /* Seconds in a day             */
        uint32  secperhr  =  3600;      /* Seconds in an hour           */
        uint32  secpermin =    60;      /* Seconds in a minute          */

        /* For argument '--help', emit help about the 'uptime' command  */

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^X Exit       ^J Justify    ^W Where Is   ^V Next Page  ^U UnCut Text ^T To
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```

2311104052_development-system [Running] - Oracle VirtualBox : 1

File Machine View Devices Help

Activities Terminal Tue 7 Apr, 04:05

xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/compile | xinu@xinu-develop-end: ~/xinu/co

```
    67          }
    68
    69          if (secs > 0) {
    70                  printf(" %d second(s) ", secs);
    71          }
    72          printf("\n");
    73
    74          return 0;
    75  }
xinu@xinu-develop-end:~/xinu$ nano ./shell/xsh_uptime.c
Use "fg" to return to nano.

[1]+  Stopped                 nano ./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ nano ./shell/xsh_uptime.c
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make
Rebuilding the .defs file
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdli
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat versior
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$
```

Right Ctrl

Activities Terminal Tue 7 Apr, 04:09

xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

```
          [0x00100000 to 0x005FBFFF]
    115625 bytes of Xinu code.
          [0x00100000 to 0x0011C3A8]
    148488 bytes of data.
          [0x0011FEC0 to 0x001442C7]

Obtained IP address  10.0.3.15   (0x0a00030f)



--------------------Izzaty zahara Br Barus--------------------
   __   __   _____   _  _   _   _
   \ \  / /  |__ __|  | \ | |  | |  | |
    \ \/ /     | |    |  \| |  | |  | |
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   / /  \ \   |     |  | | \ |  \  --  /
   --    --   -----    -   -    ----
--------------------2311104052--------------------


Welcome to Xinu!


xsh $ uptime
Xinu has been up  17 second(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```